ISOLINGO — VOCAB SCHEMA SPEC (CANONICAL STORAGE MODEL)

PURPOSE

Define the canonical vocabulary schema for Isolingo going forward.

This document establishes the difference between:
1. CANONICAL STORAGE
2. DISPLAY / GENERATION OUTPUT

This schema is intended to support:
- V0
- V1
- future grammar expansion
- user-added words
- determiner / article generation
- possessive generation

SECTION 1 — CORE PRINCIPLE

The canonical stored noun must always be the BARE LEMMA.

Do NOT store:
- articles
- determiners
- possessive pronouns
- other context-specific forms

Examples:

CORRECT CANONICAL STORAGE
- German: Hund
- French: homme
- Spanish: hombre

INCORRECT CANONICAL STORAGE
- der Hund
- l’homme
- el hombre

RATIONALE

The noun itself is the stable lexical unit.
Articles, determiners, and possessive pronouns change depending on:
- language
- gender
- number
- card type
- grammatical context

Therefore:

CANONICAL STORAGE = bare lexical item
DISPLAY / VALIDATION = generated or mapped separately

SECTION 2 — HARD RULE

For ALL languages:

lemma = bare lemma only

This applies to:
- nouns
- verbs
- adjectives
- adverbs

Examples:

NOUNS
- Hund
- homme
- hombre

VERBS
- gehen
- aller
- ir

ADJECTIVES
- gut
- bon
- bueno

ADVERBS
- heute
- aujourd’hui
- hoy

SECTION 3 — CANONICAL MASTER SCHEMA

Every vocab item must use this schema:

```
{
  id,
  targetLanguage,
  english,
  senseLabel,
  lemma,
  partOfSpeech,
  gender,
  plural,
  category,
  competencies
}
```

FIELD DEFINITIONS

1. id
- unique string identifier
- language-prefixed
- example:
  - de_hund
  - fr_homme
  - es_hombre

2. targetLanguage
- string
- allowed values:
  - "de"
  - "fr"
  - "es"
  - later expansion allowed

3. english
- primary English gloss

- string
- example:
  - "dog"
  - "man"
  - "to go"

4. senseLabel
- disambiguation label for homographs / near-homographs / multi-sense items
- string or null
- examples:
  - "financial institution"
  - "bench"
  - null

5. lemma
- canonical stored form
- ALWAYS bare lemma
- no article
- no determiner
- no possessive
- no inflected phrase

6. partOfSpeech
- string
- allowed values for current system:
  - "noun"
  - "verb"
  - "adjective"
  - "adverb"

7. gender
- required for nouns where applicable
- null for non-nouns
- allowed values:
  - "masculine"
  - "feminine"
  - "neuter"
  - null

8. plural
- noun plural form when applicable
- null for non-nouns
- string or null

9. category
- semantic category label
- required
- examples:
  - "people"
  - "food"
  - "home"
  - "travel"
  - "time"
  - "animals"
  - "work"
  - "objects"
  - "nature"
  - "general"

10. competencies
- array of strings
- determines which systems/cards may use the word

Current rules:
- noun → ["vocab","gender"]
- non-noun → ["vocab"]

SECTION 4 — REQUIRED VALUE RULES BY PART OF SPEECH

NOUNS

Required:
- id
- targetLanguage
- english
- lemma
- partOfSpeech = "noun"
- gender
- plural
- category
- competencies = ["vocab","gender"]

Allowed:
- senseLabel = string or null

Example:

```
{
  id: "es_hombre",
  targetLanguage: "es",
  english: "man",
  senseLabel: null,
  lemma: "hombre",
  partOfSpeech: "noun",
  gender: "masculine",
  plural: "hombres",
  category: "people",
  competencies: ["vocab","gender"]
}
```

VERBS

Required:
- id
- targetLanguage
- english
- lemma
- partOfSpeech = "verb"
- category
- competencies = ["vocab"]

Must use:
- gender = null
- plural = null

Example:

```
{
  id: "de_gehen",
  targetLanguage: "de",
  english: "to go",
  senseLabel: null,
  lemma: "gehen",
  partOfSpeech: "verb",
  gender: null,
  plural: null,
  category: "verbs",
  competencies: ["vocab"]
}
```

ADJECTIVES

Required:
- id
- targetLanguage
- english
- lemma
- partOfSpeech = "adjective"
- category
- competencies = ["vocab"]

Must use:
- gender = null
- plural = null

ADVERBS

Required:
- id
- targetLanguage
- english
- lemma
- partOfSpeech = "adverb"
- category
- competencies = ["vocab"]

Must use:
- gender = null
- plural = null

SECTION 5 — DISPLAY / GENERATION LAYER

The canonical master schema does NOT store article-bearing noun phrases inside lemma.

Instead, card-generation logic must produce display forms separately.

This applies especially to:
- French
- Spanish
- later German determiner-based cards
- possessive-based cards

Examples:

Canonical noun:
- lemma: hombre

Possible generated display outputs:
- el hombre
- un hombre
- mi hombre

Canonical noun:
- lemma: femme

Possible generated display outputs:
- la femme
- une femme
- ma femme

Canonical noun:
- lemma: Hund

Possible generated display outputs:
- der Hund
- ein Hund
- mein Hund

RULE

Display forms are NOT the canonical stored noun.

They are:
- generated
- mapped
- or stored separately from lemma if needed later

But they must never replace the bare lemma in the master dataset.

SECTION 6 — USER-ADDED WORD RULE

All user-added words must conform to the same canonical schema as the master dataset.

This means:

When a user adds a noun, they must enter:

- bare lemma only
- no article
- no determiner

Examples of valid noun input:
- Hund
- homme
- hombre

Examples of invalid noun input:
- der Hund
- l’homme
- el hombre

System must then map the user input into the full schema:
- generate id
- assign targetLanguage
- assign competencies
- preserve bare lemma
- store gender / plural where applicable

SECTION 7 — LANGUAGE-SPECIFIC STORAGE RULES

GERMAN

Nouns:
- bare lemma only
- no der / die / das in lemma

French:
- bare lemma only
- no le / la / l’ in lemma

Spanish:
- bare lemma only
- no el / la in lemma

This rule applies even if the card later displays:
- a definite article
- an indefinite article
- a possessive pronoun
- another determiner

SECTION 8 — SCHEMA EXAMPLES

GERMAN NOUN

```
{
  id: "de_hund",
  targetLanguage: "de",
  english: "dog",
  senseLabel: null,
  lemma: "Hund",
  partOfSpeech: "noun",
  gender: "masculine",
  plural: "Hunde",
  category: "animals",
  competencies: ["vocab","gender"]
}
```

FRENCH NOUN

```
{
  id: "fr_homme",
  targetLanguage: "fr",
  english: "man",
  senseLabel: null,
  lemma: "homme",
  partOfSpeech: "noun",
  gender: "masculine",
  plural: "hommes",
  category: "people",
  competencies: ["vocab","gender"]
}
```

SPANISH NOUN

```
{
  id: "es_hombre",
  targetLanguage: "es",
  english: "man",
  senseLabel: null,
  lemma: "hombre",
  partOfSpeech: "noun",
```

```
  gender: "masculine",
  plural: "hombres",
  category: "people",
  competencies: ["vocab","gender"]
}
```

GERMAN VERB

```
{
  id: "de_gehen",
  targetLanguage: "de",
  english: "to go",
  senseLabel: null,
  lemma: "gehen",
  partOfSpeech: "verb",
  gender: null,
  plural: null,
  category: "verbs",
  competencies: ["vocab"]
}
```

FRENCH ADJECTIVE

```
{
  id: "fr_bon",
  targetLanguage: "fr",
  english: "good",
  senseLabel: null,
  lemma: "bon",
  partOfSpeech: "adjective",
  gender: null,
  plural: null,
  category: "adjectives",
  competencies: ["vocab"]
}
```

SPANISH ADVERB

```
{
  id: "es_hoy",
  targetLanguage: "es",
  english: "today",
  senseLabel: null,
  lemma: "hoy",
```

```
  partOfSpeech: "adverb",
  gender: null,
  plural: null,
  category: "adverbs",
  competencies: ["vocab"]
}
```

SECTION 9 — CARD SYSTEM IMPLICATION

Card systems must treat:

lemma
as:
- the canonical lexical item

and must treat:

article / determiner / possessive form
as:
- generated card content
- not canonical storage

This is necessary because the same noun may appear on different cards as:

Spanish:
- el hombre
- un hombre
- mi hombre

French:
- l’homme
- un homme
- mon homme

German:
- der Hund
- ein Hund
- mein Hund

The stored noun must remain:
- hombre
- homme

- Hund

SECTION 10 — FINAL LOCKED RULE

From this point forward:

1. All new vocab entries must use bare lemma storage
2. No article-bearing noun phrase may be stored in lemma
3. Existing French and Spanish noun entries that include articles must be normalized later
4. Card-generation logic may display article-bearing forms, but the master dataset must store only the bare noun

END DOCUMENT